

File permissions in Linux

Project description

Through Linux commands, I will investigate the permissions of users, groups and others for files and directories within a projects directory, as well as altering them where needed.

Check file and directory details

```
researcher2@f3cbc42ed433:~$ pwd
/home/researcher2
researcher2@f3cbc42ed433:~$ cd projects
researcher2@f3cbc42ed433:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$
```

Describe the permissions string

For a string drwxrwxrwx, the first letter refers to a directory, r = read, w = write, x = execute/access the file. The first set of rwx is for user permissions, the second for group permissions and the last for others' permissions. The presence of d in the first space means it is a directory, where a hyphen means it is a file. The presence of an r/w/x means that permission is granted to its respective user type, whereas a hyphen would mean it is not granted.

Change file permissions

```
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$ chmod o-w project_k.txt
researcher2@f3cbc42ed433:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$
```

```
researcher2@f3cbc42ed433:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
```

Change file permissions on a hidden file

```
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 20:45 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 21:03 ..
-rw--w---- 1 researcher2 research_team  46 Aug 29 20:45 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@f3cbc42ed433:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 20:45 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 21:03 ..
-r--r----- 1 researcher2 research_team  46 Aug 29 20:45 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$
```

Change directory permissions

```
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 20:45 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 29 21:03 ..
-r--r----- 1 researcher2 research_team 46 Aug 29 20:45 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$ chmod g-x drafts
researcher2@f3cbc42ed433:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Aug 29 20:45 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 29 20:45 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 29 20:45 project_t.txt
researcher2@f3cbc42ed433:~/projects$
```

Summary

I have used the bash shell commands `cd`, `ls` and `chmod` to navigate to a directory, discern file and directory permissions, and change these permissions.